

1, Consider this Hidden Markov model for the Dishonest Casino problem.

A casino has two dice:

- Fair die

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$$

- Loaded die

$$P(1) = P(2) = P(3) = P(4) = P(5) = 1/10; \quad P(6) = 1/2$$

The Transition probability (switching between fair and loaded die) is 0.05

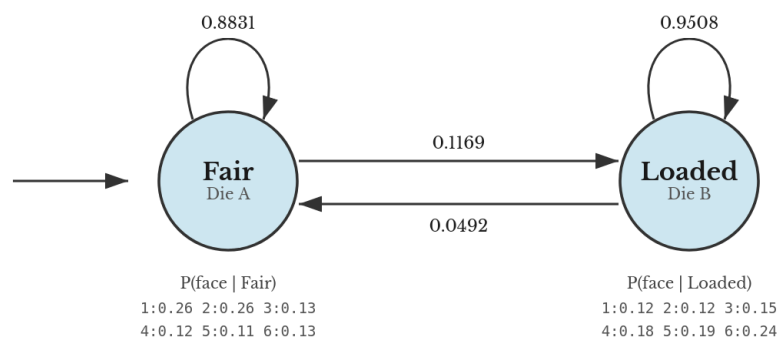
1. 1, We have a series of die rolling outcomes "61226351241253662512" what is the probability that first half are generated by the fair die, and second half are generated by the loaded die?

- $P = 8.21 \times 10^{-19}$

1.2, We have a series of die rolling outcomes (*HMM_DiceFaces1000.txt*, download from course website), Try your best to guess which outcomes are generated by the fair die, and which are by the loaded die.

- About 650 rolls are generated by the fair die.
- About 350 rolls are generated by the loaded die.

2.1 Draw the FSA (finite state automata) for this Hidden Markov model. With all the probabilities.



2.2 Try to predict which outcomes are generated by the fair die, and which are by the loaded die.

(Already did in-class for the presentation - Group 4)